

Problem Set 1a: Space Cows - Answers

1. Results from `compare_cow_transport_algorithms()`

Results:

Greedy Algorithm: 6 trips in 0.000015 seconds

Brute Force Algorithm: 5 trips in 0.269619 seconds

Greedy trips:

[['Betsy'], ['Henrietta'], ['Herman', 'Maggie'], ['Oreo', 'Moo Moo'], ['Millie', 'Milkshake', 'Lola'], ['Florence']]

Brute Force trips:

[['Henrietta'], ['Florence', 'Oreo', 'Lola'], ['Milkshake', 'Millie', 'Maggie'], ['Moo Moo', 'Herman'], ['Betsy']]

Which runs faster? The greedy algorithm runs significantly faster—approximately 18,000 times faster (0.000015 seconds vs 0.269619 seconds).

Why? The greedy algorithm has a time complexity of $O(n \log n)$ due to sorting the cows once. It then makes a single pass through the sorted list for each trip. In contrast, the brute force algorithm has exponential time complexity $O(2^n)$ because it must enumerate all possible partitions of the cows. With 10 cows, this means checking millions of partition combinations. As the number of cows increases, the brute force algorithm becomes prohibitively slow while the greedy algorithm remains efficient.

2. Does the greedy algorithm return the optimal solution?

No, the greedy algorithm does not return the optimal solution.

As shown in the results above, the greedy algorithm returned 6 trips while the brute force algorithm found an optimal solution with only 5 trips. The greedy algorithm is a heuristic that makes locally optimal choices (always packing the heaviest available cow) without considering the global problem structure. This "greedy choice property" can lead to suboptimal solutions because it doesn't account for how smaller cows might pack better together.

In this case, the greedy approach placed heavy cows like Betsy (9) and Henrietta (9) in their own trips, while a more optimal arrangement groups them with lighter cows in ways that minimize total trips needed.

3. Does the brute force algorithm return the optimal solution?

Yes, the brute force algorithm returns the optimal solution.

The brute force algorithm is guaranteed to find the optimal solution because it exhaustively searches all possible partitions of the cows and selects the one with the fewest trips that respects the weight limit constraint. By examining every valid partition, it cannot miss the optimal solution. The algorithm's correctness comes from its completeness—it considers all possibilities.

However, this guarantee comes at a computational cost: exponential time complexity. The brute force approach is only practical for small problem sizes (in this case, 10 cows). For larger datasets, the exponential growth in computation time makes it infeasible, which is why heuristic algorithms like the

greedy approach are often used in practice despite not guaranteeing optimality.